Ultimate Data Science Challenge

Part 1 - Exploratory data analysis

The login file contains 93,142 timestamps spanning 1970-01-01 20:12 through 1970-04-13 18:57. I aggregated the events into 15-minute buckets, producing 9,788 intervals. The median interval contains 7 logins, while the mean is 9.52, indicating a right-skewed distribution driven by occasional high-demand bursts.

Metric	Value
Total raw login records	93,142
15-minute intervals	9,788
Mean logins per 15 minutes	9.52
Median logins per 15 minutes	7
Peak interval count	73
Peak interval timestamp	1970-03-01 04:30
First day total (partial day)	112
Last day total (partial day)	395
Average full-day total	917.18

Demand patterns

The time series is strongly cyclical rather than trendless noise. Activity rises from Monday through Saturday, then eases slightly on Sunday. Average login volume by day increases from 6.21 per 15-minute interval on Monday to 13.46 on Saturday, with Sunday still elevated at 12.62.

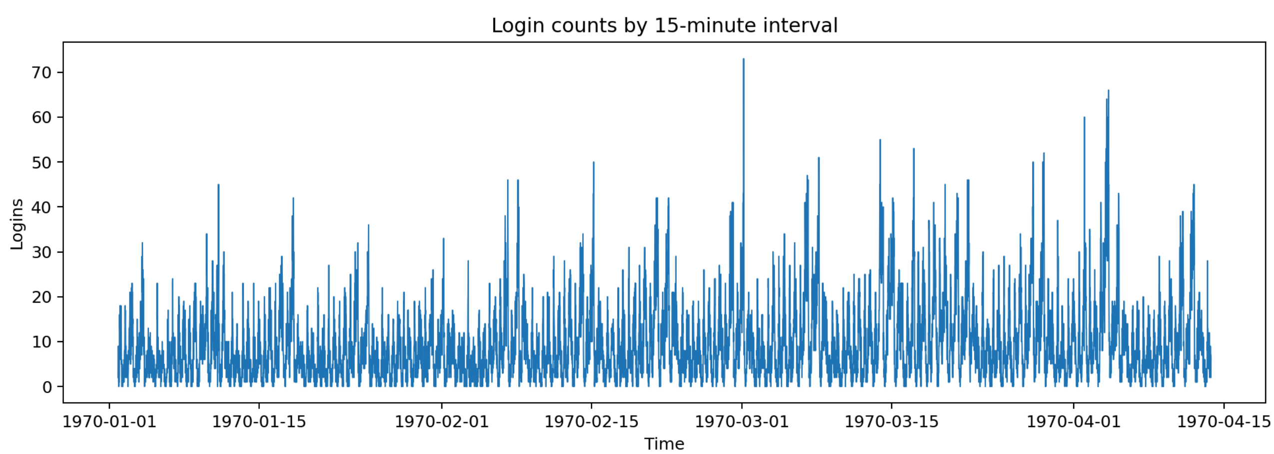

Figure 1. Raw login counts aggregated into 15-minute intervals.

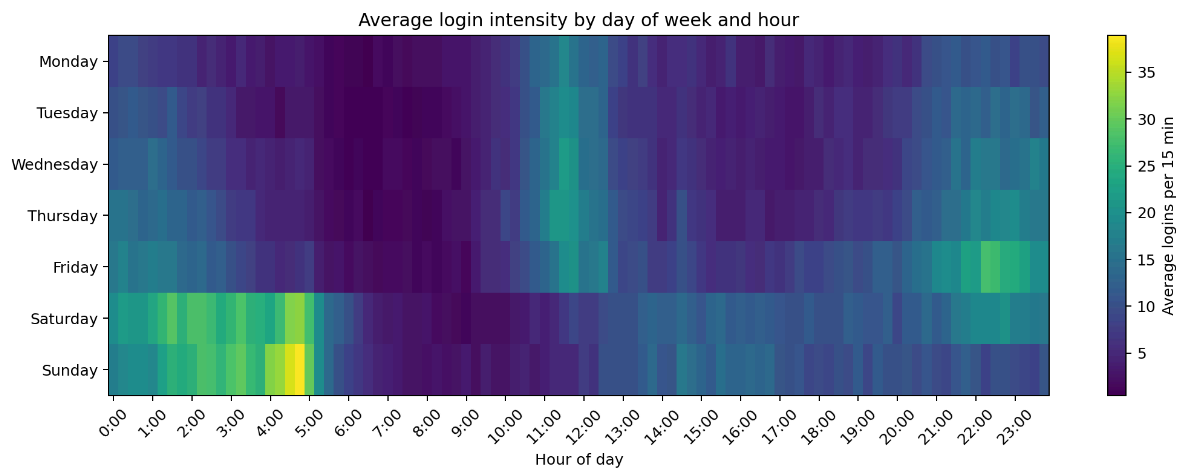

Figure 2. Strongest demand appears on weekend nights and early mornings.

Main findings

- Weekday demand has a midday peak centered around 11:30 AM, averaging about 19.96 logins per 15 minutes.
- Weekend demand peaks around 4:45 AM, averaging about 35.63 logins per 15 minutes, which is substantially higher than the weekday peak.
- Friday night transitions into the Saturday early-morning surge, and Saturday night transitions into the Sunday early-morning surge.
- Demand is lowest during early weekday mornings after the overnight period and before the midday build.
- The series does not show an obvious long-run upward or downward trend over the sample; the repeated intraday and weekly cycles dominate.

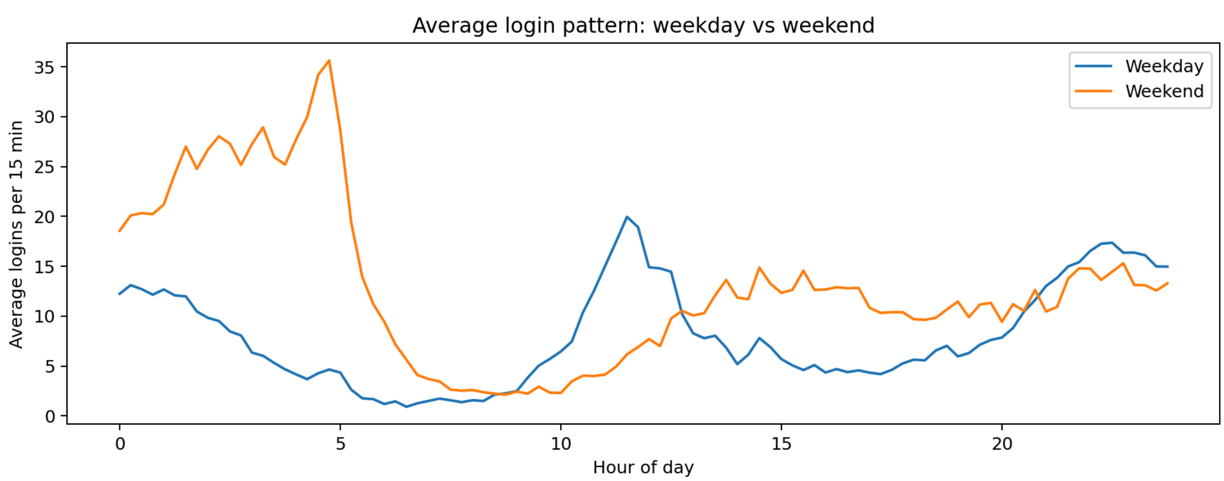

Figure 3. Weekday and weekend curves differ sharply, especially overnight.

Data quality notes

The first and last calendar days are incomplete because the series begins in the evening of January 1 and ends before 7 PM on April 13. Those boundary days should be excluded from any day-over-day operational comparison. Aside from that boundary issue, the timestamps appear internally consistent, with no impossible dates and no need for deduplication assumptions beyond the provided event stream.

Part 2 - Experiment and metrics design

The business goal is to increase cross-city supply by reimbursing tolls, so the success metric should capture whether drivers actually become active in both cities rather than simply driving more overall.

Recommended primary metric

Use the share of active drivers who complete at least one trip in both Gotham and Metropolis during the evaluation window. This metric directly measures the desired behavior change, is easy to explain to operations stakeholders, and is harder to misread than total trip volume because it isolates cross-city participation.

Metric type	Recommendation
Primary KPI	Percent of active drivers serving both cities during the test window
Guardrail 1	Total completed trips per active driver
Guardrail 2	Rider ETAs / fulfillment rate in each city
Guardrail 3	Driver hourly earnings net of toll reimbursements
Guardrail 4	Incremental reimbursement cost per additional dual-city driver

Experiment design

- Randomize at the driver-partner level, not the city level. Treatment drivers receive full toll reimbursement for all bridge crossings during the test period; control drivers continue under the current policy.
- Use a pre-test baseline period to stratify randomization by historical trip volume, home city, day-part mix, and prior bridge-crossing behavior if available. That keeps treatment and control balanced on the characteristics most likely to affect cross-city participation.
- Run the test long enough to cover multiple full weekday and weekend cycles. Four to six weeks is a practical minimum because driver behavior is highly day-part dependent.
- Communicate the policy clearly in the treatment group app experience so lack of awareness does not dilute the effect estimate.

Statistical test

For the primary metric, compare the treatment and control proportions with a two-sample test for proportions (equivalently, a chi-square test on a 2x2 table). A logistic regression with treatment assignment and baseline covariates can be used as a robustness check and to estimate heterogeneous effects across driver segments. For continuous guardrails such as trips per driver or earnings, use a two-sample t-test or a non-parametric alternative if the distributions are highly skewed.

Interpretation and recommendation framework

- If the treatment group shows a statistically significant and operationally meaningful lift in the dual-city driver share, with neutral or positive rider service levels and acceptable unit economics, scale the reimbursement program.
- If the dual-city share rises but the incremental cost per added dual-city driver is too high, keep the concept but target it narrowly to high-need day-parts rather than all crossings.
- If the treatment effect is weak or insignificant, do not roll out broadly; instead test more targeted incentives such as time-of-day reimbursements, minimum-crossing bonuses, or guaranteed earnings windows.

- Key caveat: reimbursement may change who is online, where drivers position themselves, and how long they stay active. The analysis should therefore include service quality and earnings guardrails, not just the primary KPI.

Part 3 - Predictive modeling

The rider file contains 50,000 January 2014 signups. Following the standard assumption for this challenge, I labeled a user as retained if their last trip date was on or after 2014-06-01, which approximates being active in the 30 days prior to the July 1 pull date.

Metric	Value
Observed users	50,000
Estimated retained users	18,804
Estimated retention rate	37.61%
Missing avg_rating_of_driver	16.24%
Missing phone	0.79%
Missing avg_rating_by_driver	0.40%

Exploratory findings

- Retention differs sharply by city: King's Landing 62.8%, Winterfell 35.2%, Astapor 25.6%.
- iPhone users retain at a much higher rate than Android users: 44.9% versus 20.9%.
- Users who took Ultimate Black in their first 30 days retain at 50.4% versus 29.9% for non-users.
- Early engagement matters: retention climbs from 27.3% for riders with 1 to 2 trips in month one to 73.6% for riders with 11 or more trips.

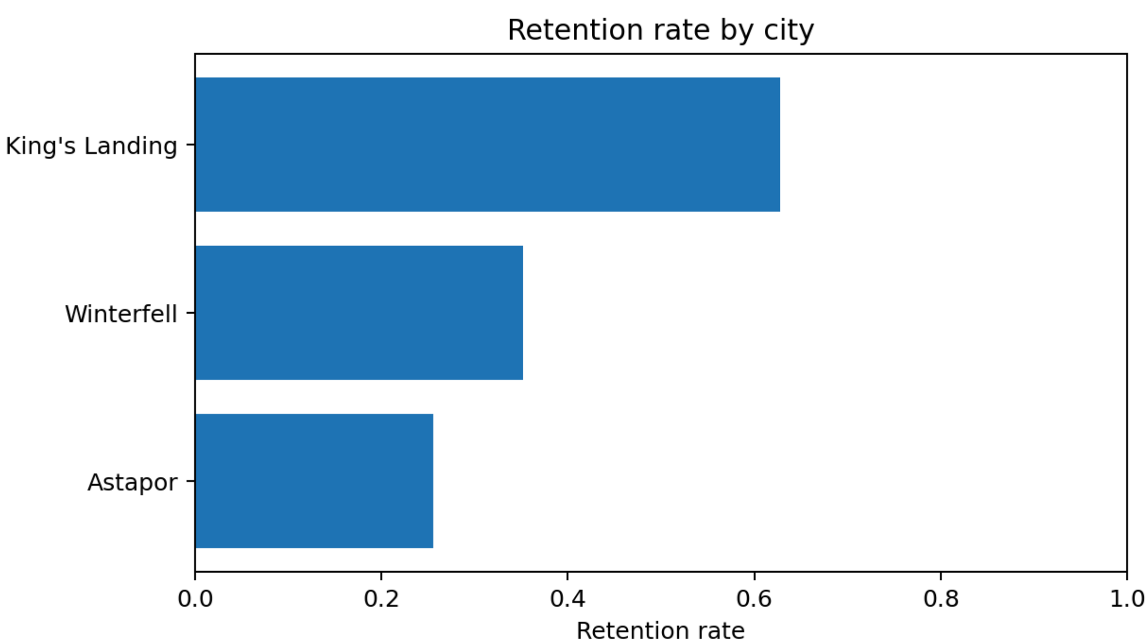

Figure 4. City-level retention varies substantially.

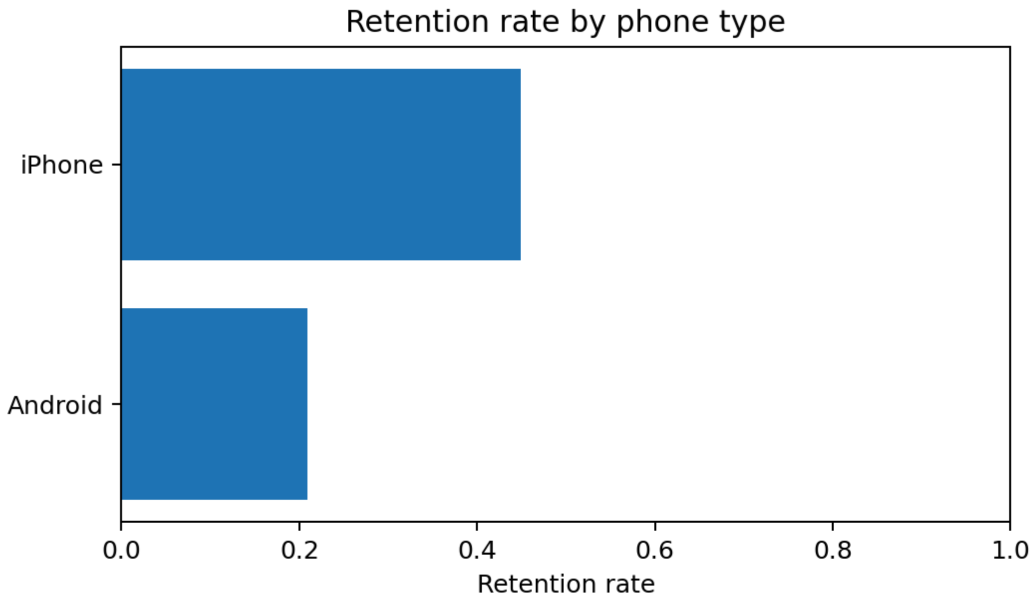

Figure 5. Phone type is also strongly associated with retention.

Modeling approach

I compared an imputed, one-hot encoded logistic regression baseline with a random forest classifier. I excluded last_trip_date from the features because it directly defines the target and would create leakage. Missing numeric values were median-imputed, categorical values were imputed with the most frequent class, and the data was split into stratified train and holdout sets.

Model	ROC-AUC	Avg precision	Accuracy	F1
Logistic regression	0.754	0.659	0.717	0.565
Random forest	0.850	0.784	0.782	0.689

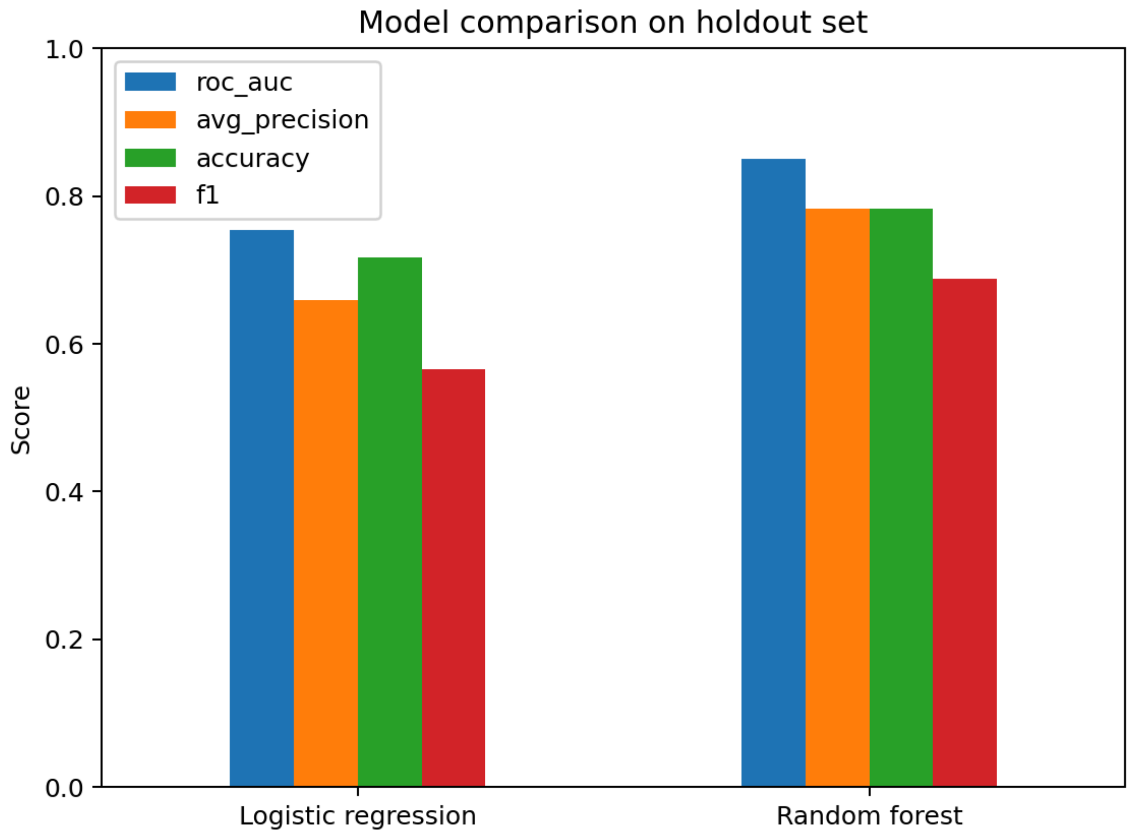

Figure 6. Random forest is the stronger predictive model on the holdout set.

Why random forest wins here

- It captures nonlinear effects such as sharp improvements in retention after a rider crosses certain early-usage thresholds.
- It handles interactions naturally, for example between city, phone type, and early engagement behavior.
- It performed materially better on discrimination metrics while still remaining operationally usable.

Operational recommendations

- Focus activation efforts on riders with weak first-30-day engagement. The month-one trip count is one of the clearest practical signals, so onboarding and reactivation nudges should be concentrated early.
- Target lower-retention segments by city. Astapor and Winterfell appear to need different retention playbooks than King's Landing.
- Promote premium or loyalty-style experiences selectively. Ultimate Black usage is associated with meaningfully higher retention, which suggests premium engagement may deepen stickiness for some users.
- Use predicted retention scores to prioritize lifecycle messaging, incentives, and CRM budget. High-risk users can receive early ride credits or commute reminders, while high-likelihood retainers may not need subsidy.